

UFC EDGE AUDIT

Could a model built only from information available before each fight have out-priced the betting market? We rebuilt every prediction as if it were made on the day, froze the design, and then scored years it had never seen.

 Blue corner: the engine  Red corner: the betting market

THE SHORT ANSWER

A small, repeatable edge — biggest when lines first open

On the locked holdout (Jan 2025 to Sep 2026, 878 fights) the engine's probabilities beat the closing line by +0.0064 log-loss per fight and the opening line by +0.0153. That is modest in accuracy terms — both sides pick about the same winners — but it is where money is made: the engine is better at saying *how likely*, not just *who*.

WINNER-PICKING, UNSEEN FIGHTS 69.9% engine with market vs 69.8% closing favourite; stats alone 67.2%	EARLY BETS, 2021-SEP 2026 +12.1% return on stake over 1,418 bets at opening prices	LINE MOVED OUR WAY +11.4% average closing-line value of those early bets
LATE BETS, 5%-MARGIN BOOK +9.6% over 324 bets — positive, but not proven		

DID IT BEAT THE MARKET?

Probability quality, fight by fight

Log loss rewards confident correct probabilities and punishes confident wrong ones; lower is better. “Gain” is the average per-fight improvement over the market, with a bootstrap confidence interval. The development years were used to choose the design and are shown only for context.

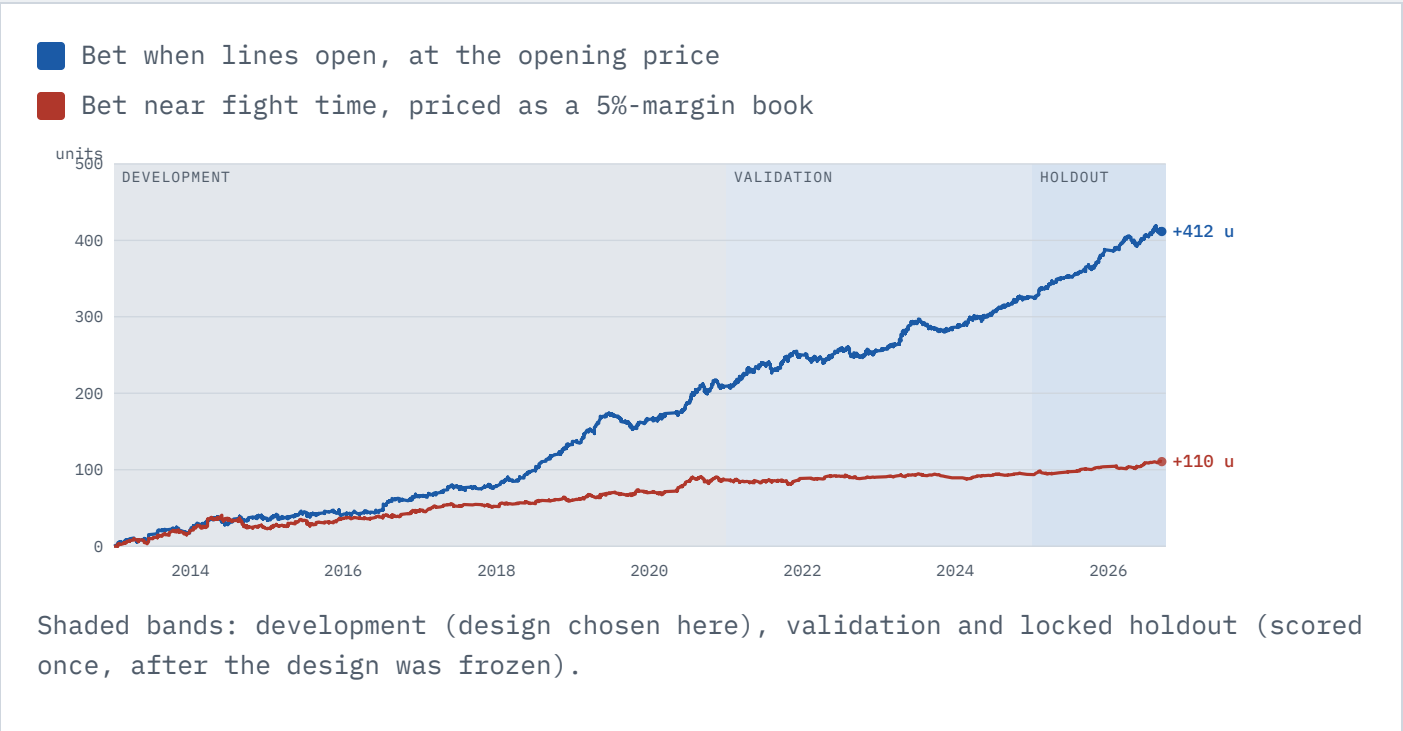
PERIOD	CLOSING LINE	ENGINE	GAIN VS CLOSE	OPENING LINE	ENGINE (OPEN)	GAIN VS OPEN	ACCUR
2011–2020 Development · 4,233 fights · selection-biased	0.6114	0.6059	+0.0055 95% CI +0.0033 to +0.0077	0.6274	0.6159	+0.0115 95% CI +0.0079 to +0.0149	66.0%
2021–2024 Validation · 1,969 fights · untouched	0.6018	0.5992	+0.0026 95% CI -0.0001 to +0.0050	0.6246	0.6147	+0.0099 95% CI +0.0046 to +0.0157	67.9%

PERIOD	CLOSING LINE	ENGINE	GAIN VS CLOSE	OPENING LINE	ENGINE (OPEN)	GAIN VS OPEN	ACCUR
2025-Sep 2026	0.5779	0.5716	+0.0064	0.5973	0.5820	+0.0153	69.8%
Locked holdout · 878 fights · untouched			95% CI +0.0023 to +0.0105	95% CI +0.0072 to +0.0229			

WOULD IT HAVE MADE MONEY?

Cumulative units, one disciplined rule the whole way

Rule: bet only when expected value exceeds 3%, stake 1/8 Kelly, never more than 1.5 units (1 unit = 1% of bankroll), one bet per fight, draws lose, no-contests refund. Every probability behind every bet came from a model trained only on earlier fights.



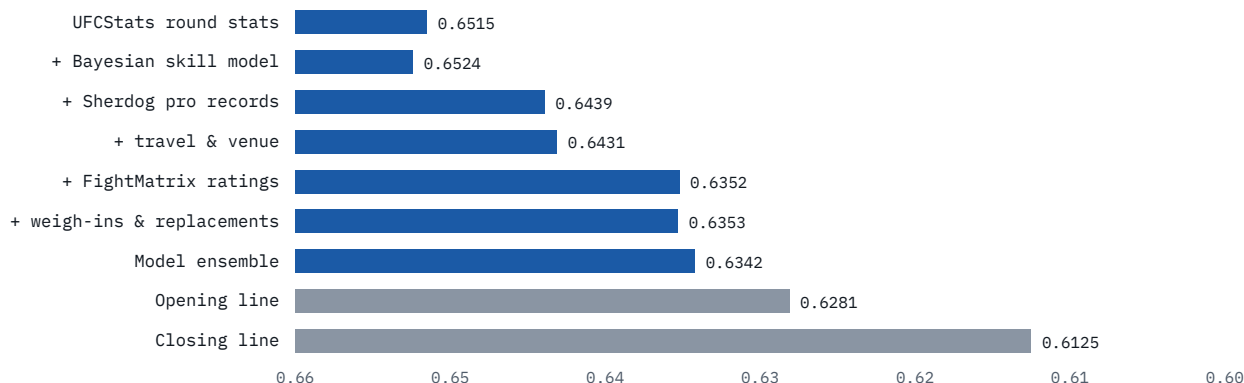
SETTING	BETS	STAKED	PROFIT	RETURN	WORST DIP	YEARS UP
Bet when lines open (opening price) 2021-2024	1,012	1,194 u	+117 u	+9.8% 90% CI +5.2% to +14.7%	-17 u	4 of 4

SETTING	BETS	STAKED	PROFIT	RETURN	WORST DIP	YEARS UP
Bet when lines open (opening price) 2025-Sep 2026	406	479 u	+85 u	+17.8% 90% CI +10.4% to +25.1%	-12 u	2 of 2
Bet when lines open (opening price) 2021-Sep 2026	1,418	1,673 u	+203 u	+12.1% 90% CI +8.1% to +16.2%	-18 u	6 of 6
Bet near fight time (5%-margin book) 2021-2024	239	178 u	+7 u	+4.0% 90% CI -6.8% to +14.4%	-8 u	3 of 4
Bet near fight time (5%-margin book) 2025-Sep 2026	85	67 u	+17 u	+24.7% 90% CI +7.2% to +40.9%	-4 u	2 of 2
Bet near fight time (5%-margin book) 2021-Sep 2026	324	246 u	+24 u	+9.6% 90% CI +0.4% to +18.9%	-7 u	5 of 6

Read this before staking anything. Opening lines are the softest prices but often carry low limits, so the early-bet result is real but not scalable. The late-bet result is positive in the holdout yet was not statistically significant in validation, so treat it as unproven. The worst historical dip is a floor, not a ceiling.

WHICH INFORMATION CARRIES SIGNAL?

Records from outside the UFC matter more than cage statistics



Market-free model, development years, log loss as each source is added (longer bar = better). Grey bars are the betting lines for scale.

UFCStats striking and grappling numbers alone reach 0.652. The biggest single jumps came from each fighter’s full professional record on Sherdog (pre-UFC fights, a global Elo across every promotion) and from FightMatrix’s pre-fight ratings. A Bayesian 14-skill model of opponent-adjusted offence and defence was elegant but added nothing the engineered statistics didn’t already carry. Age was the most important single physical variable; reach, height and stance barely moved the needle.

After controlling for the closing line, the corrections that replicated were consistent with an intuitive story: the market underweights *defence* (strikes absorbed, striking defence) and wrestling pressure (takedown attempts), and slightly overreacts to late line movement.

CONVENTIONAL WISDOM VS THE DATA

Patterns found before 2021, re-tested on 2021–2026

Each was a flat one-unit bet at the average closing price. Most historical “angles” disappeared once the calendar moved on — which is exactly why none of them is allowed to drive the engine on its own.

ANGLE	BEFORE 2021	2021- SEP 2026	
Main-card slight underdogs (37.5-45%)	+13.0% 582 bets	-4.9% 307 bets · 2/6 years up	Faded: +13% before 2021, nega
Big underdogs (market ≤ 20%)	-21.1% 422 bets	-44.3% 305 bets · 0/6 years up	Holds: longshots are overbet
Big favourites (market ≥ 80%)	-0.1% 422 bets	+3.2% 305 bets · 5/6 years up	Mild: small profit, 5 of 6 re
Against UFC debutants vs 3+ fight veterans	backing them: -12.7% 531 bets	-9.4% 278 bets · 2/6 years up	Faded
Against short-notice replacements	backing them: -13.5% 522 bets	-6.4% 361 bets · 1/6 years up	Faded
Against fighters who missed weight	backing them: -32.5% 98 bets	+8.7% 163 bets · 3/6 years up	Same direction, too few bet
Younger fighter by 5+ years	+2.7% 1640 bets	+0.9% 1030 bets · 3/6 years up	Tiny edge, no
Home-country fighter vs foreigner	not tested	-9.2% 1338 bets · 1/6	Recent: home fighters ove

ANGLE

BEFORE 2021

2021-
SEP
2026years
up

Two more myths the data rejected: fighters who missed weight won only 40% of their bouts (n = 259), not the 55–60% often quoted, and the eye-catching “5’10” beats 6’1” only 26% of the time” cell shrank to 40% on later fights — small-sample noise.

WHERE IT SUCCEEDS AND FAILS

Strong on known quantities, no edge on debutants

By UFC experience (fewer fights of the two)

GROUP	FIGHTS	MARKET	ENGINE	GAIN	ACC. MKT / ENG
both debut	69	0.5342	0.5395	-0.0053	76.8% / 76.8%
one debutant	460	0.5580	0.5583	-0.0003	70.9% / 70.4%
min 1-2 UFC fights	748	0.6018	0.5952	+0.0066	66.7% / 67.0%
min 3-5	686	0.5954	0.5911	+0.0042	69.8% / 70.1%
min 6+	884	0.6111	0.6073	+0.0037	67.1% / 68.6%

By market favourite price

GROUP	FIGHTS	MARKET	ENGINE	GAIN	ACC. MKT / ENG
0.50-0.60	879	0.6812	0.6762	+0.0049	56.8% / 58.5%
0.60-0.70	1,033	0.6454	0.6456	-0.0002	65.6% / 65.6%
0.70-0.80	631	0.5190	0.5140	+0.0050	78.8% / 78.8%
0.80-1.01	304	0.3270	0.3159	+0.0111	90.8% / 90.8%

By division

	GROUP	FIGHTS	MARKET	ENGINE	GAIN	ACC. MKT / ENG
	Bantamweight	423	0.5891	0.5800	+0.0091	70.0% / 71.6%
	Lightweight	375	0.6093	0.6031	+0.0062	68.8% / 69.6%
	Featherweight	354	0.6046	0.5971	+0.0074	65.3% / 66.1%
	Heavyweight	205	0.5550	0.5504	+0.0046	72.7% / 73.7%
	Light Heavyweight	201	0.6269	0.6384	-0.0116	62.7% / 62.7%
	women	479	0.5933	0.5890	+0.0044	69.1% / 70.1%

When engine and market pick different winners

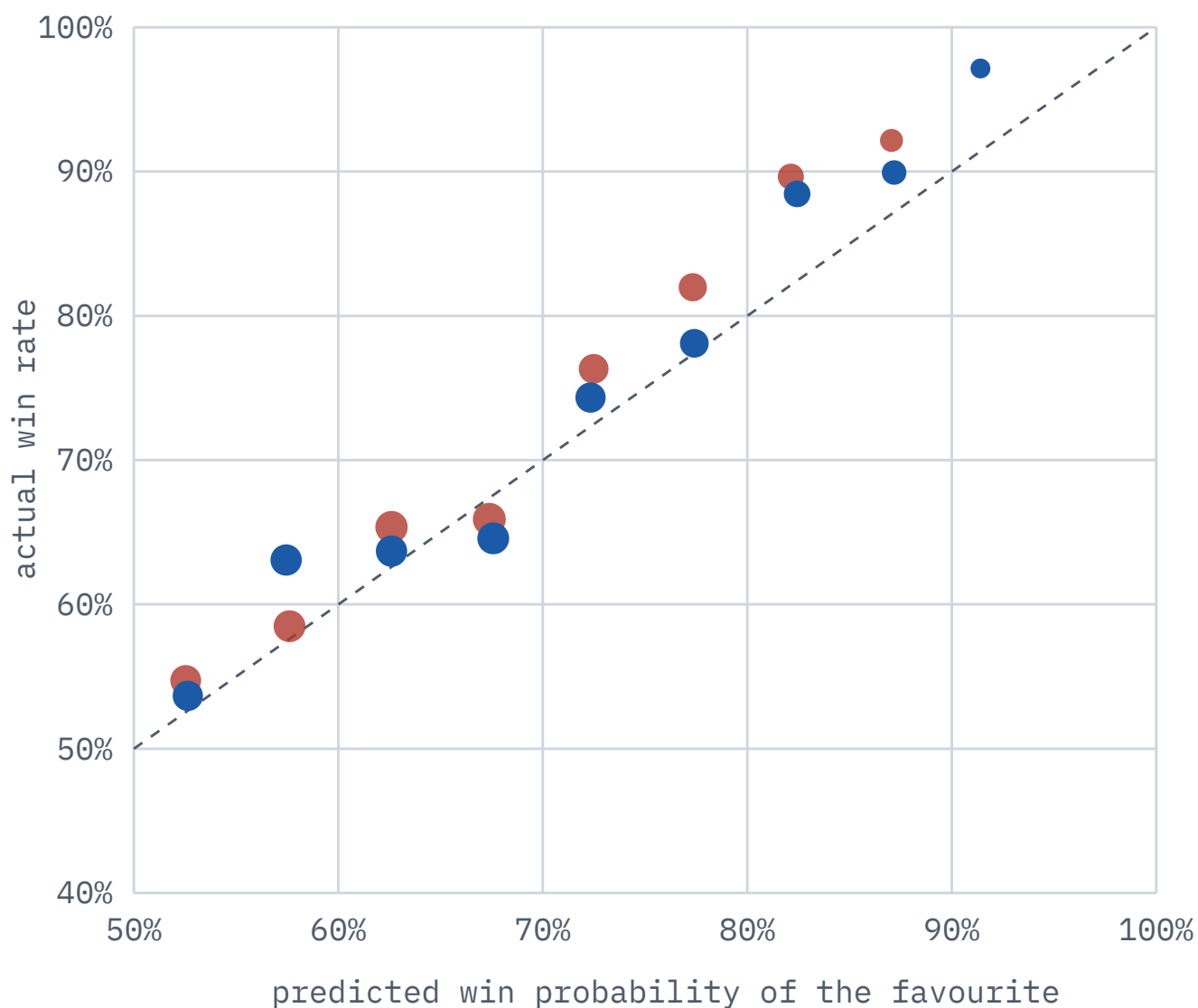
	GROUP	FIGHTS	MARKET	ENGINE	GAIN	ACC. MKT / ENG
	engine & market agree	2,764	0.5912	0.5879	+0.0033	69.3% / 69.3%
	engine disagrees with market	83	0.7022	0.6834	+0.0188	41.0% / 59.0%

All 2021–Sep 2026, log loss (lower is better). The market is sharper than the engine on debutants — it knows regional reputations and camp news the data doesn’t. Light heavyweight is the one division where the engine loses ground. The biggest misses were genuine upsets priced above 80% by both sides (Nunes–Peña, Shevchenko–Grasso, Adesanya–Strickland): the fights the engine should be least confident about are the ones nobody can call.

HOW WELL CALIBRATED IS IT?

When it says 70%, it wins about 70%

■ Engine ■ Closing line



2021-Sep 2026. Dot size = fights in the bin; dashed line = perfect calibration.

Both sides sit close to the diagonal. The one persistent market flaw is at the extremes: favourites priced above 80% win more often than their price implies, and longshots under 20% are overbet — backing them lost 44% of stake since 2021. The engine inherits the line's strengths and leans into that tilt, which is where a good share of its gain comes from.

USING IT

One command before each card

```
python -m ufc.predict --update
```

It refreshes every source, builds features for the next 30 days of cards, and prints the pick, the engine’s probability, the market’s, the edge at the average price, and a stake in units. Example output from 14 Sep 2026 for UFC 331 (odds move; these are illustrations, not current advice):

BOUT	PICK	ENGINE	MARKET	BET ON	Average Price
Joshua Van vs Alexandre Pantoja Flyweight	Joshua Van	61%	54%	Joshua Van	-133
Arman Tsarukyan vs Mauricio Ruffy Lightweight	Arman Tsarukyan	75%	73%	-	-330
Patricio Freire vs Doo Ho Choi Featherweight	Doo Ho Choi	74%	69%	-	-263
Renato Carneiro vs Brian Ortega Lightweight	Renato Carneiro	69%	65%	-	-214
Alonzo Menifield vs Iwo Baraniewski Light Heavyweight	Iwo Baraniewski	70%	66%	-	-231
Gable Steveson vs Sean Sharaf Heavyweight	Gable Steveson	93%	90%	-	-1662
Marlon Vera vs Charles Jourdain Bantamweight	Charles Jourdain	73%	67%	Charles Jourdain	-236
Tai Tuivasa vs Robelis Despaigne Heavyweight	Robelis Despaigne	82%	82%	-	-636
Edmen Shahbazyan vs Bruno Ferreira Middleweight	Edmen Shahbazyan	59%	59%	-	-165
Giga Chikadze vs Joanderson Brito Featherweight	Joanderson Brito	80%	75%	-	-380

BOUT	PICK	ENGINE	MARKET	BET ON	A PRI
Casey O'Neill vs Eduarda Moura Flyweight	Casey O'Neill	60%	62%	-	-186
Ryan Gandra vs Osman Diaz Light Heavyweight	Ryan Gandra	82%	78%	-	-468
Michael Aswell vs Joo Sang Yoo Featherweight	Joo Sang Yoo	69%	70%	-	-278

- Run it as soon as lines open; that is where the tested edge was largest.
- Never take a worse price than the one shown; recheck edges on fight day.
- Keep the stake rule. Doubling units on a “sure thing” is how a real edge still goes broke.

METHOD AND SAFEGUARDS

How we tried not to fool ourselves

- **No hindsight by construction.** Features are built date by date; each fighter is snapshotted before a fight and updated after. Canary tests delete everything after mid-2016 (and shuffle inputs): all earlier features stay bit-identical.
- **Pre-registration.** Design, thresholds and success criteria were written down before scoring 2021–2024, and again before the 2025–2026 holdout. The one failed criterion (late-bet profit in validation) is reported as a failure.
- **Only future-obtainable data.** UFCStats (daily mirror), Sherdog, BestFightOdds, FightMatrix, Wikipedia and Open-Meteo can all be re-collected for next week’s card. Wikimedia pageviews were dropped when the API throttled collection.
- **Independent prices.** Our scraped odds matched two separate archives (average gap under 2.5 points, including Pinnacle). Development-period

profits were re-settled at those archives' prices and survived.

- **Controls.** A recalibrated market alone, and the market plus a random “model”, both fail to make money under the same pipeline.

Known limits: opening-line limits cap real stakes; the UFCStats mirror lags a day; a handful of overturned results are treated as final; FightMatrix can backfill forgotten regional bouts into old ratings.

Engine and research code: `ufc/` and `research/`. Every table above is regenerated by `research/30_report.py` from saved out-of-sample predictions. Past performance of a betting model is evidence, not a promise; bet only what you can afford to lose.